71. The average marks of 100 students are given to be 40. It was found later that marks of one student were 53 which were misread as 83. The corrected mean marks are

(a) 39 (b) 39.7
(c) 40 (d) 40.3

Ans : (b) The correct mean $= \frac{100\times 40+53-83}{100}$

$= \frac{4000-30}{100} = \frac{3970}{100} = 39.70$

72. If the numerator and denominator of proper fraction are increased by the same positive quantity which is greater than zero , the resulting fraction is ?

(a) always less than the original fraction
(b) always greater than the original fraction
(c) always equal to the original fraction
(d) such that noting can be claimed definitely

Ans : (b) If the numerator and denominator of proper fraction are increased by the same positive quantity which is greater than zero. The resulting fraction is always greater than the original fraction.
For example,

$\frac{1}{2} = 0.5$

$\frac{1+1}{2+1} = \frac{2}{3} \Rightarrow 0.66$

Hence $\frac{1}{2} < \frac{2}{3}$

73. A family has two children along with their parents .The average of the weights of the children and their mother is 50 Kg. The average of the weights of the children and their father is 52kg. If the weight of the father is 60kg then what is the weight of the mother ?

(a) 48 kg (b) 50 kg
(c) 52 kg (d) 54 kg

Ans : (d) $\because \frac{2c+m}{3} = 50$

$\Rightarrow 2c+m = 150$.....(i)

and $\Rightarrow \frac{2c+F}{3} = 52$

$\Rightarrow$ 2c+F= 156

$\Rightarrow$ 2c+60=156 ($\because$ F= 60 kg)

2c = 156-60

2c = 96

c = 48

Putting the value of C in equation (i)

2c+m = 150

2×48+m =150

96+m =150

m = 150-96

$\boxed{m = 54kg}$

74. An 8-digit number 4252746B leaves remainder 0 when divided by 3. How many values of B are possible?

(a) 2 (b) 3
(c) 4 (d) 6

Ans : (c) An 8-digit number 4252746 B leaves remaineder 0,
when divided by 3
$\because$ Divisibility of 3 : if the sum of digits of any number divisible by 3, the number will completely divided by 3.
then, 4 2 5 2 7 4 6 B
$\Rightarrow$ 4 + 2 + 5 + 2 + 7 + 4 + 6 + B
= 30 + B
$\therefore$ Possible value of B,
30 + 0 = 30 divisible by 3
30 + 3 = 33 divisible by 3
36 + 6 = 36 divisile by 3
30 + 9 = 39 divisible by 3
Hence,
The possible values of 13 are 4

75. Sunita cuts a sheet of paper into three pieces. Length of first piece is equal to the average of the three single digit odd prime numbers. Length of the second piece is equal to that of the first plus one-third the length of the third. The third piece is as long as the other two pieces together. The length of the original sheet of paper is-

(a) 13 units (b) 15 units
(c) 16 units (d) 30 units

Ans : (d) Let
The length of three pieces of papes is x, y and z respectively
then,

$x = \frac{3+5+7}{3}$ $x + \left(z\times\frac{1}{3}\right) = y$

$x = \frac{15}{3}$ $x + \frac{z}{3} = y$ .....(ii)

x= 5.....(i) z = x+y.....(iii)

From eg.n (ii) & (iii)

$x + \frac{2}{3} = y$ ($\because z = x + y$)

$x + \frac{(x+y)}{3} = y$

$\frac{3x+x+y}{3} = y$

4x + y = 3y

4x = 2y

$\boxed{y = 2x}$

y = 2 × 5 from eqn (i) $\{\because x = 5\}$

$\boxed{y = 10}$

$\because z = x + y$
z = 5 + 10
$\boxed{z = 15}$
$\therefore$ The length of the original sheet of paper
= x + y + z = 5 + 10 + 15 = 30 units

76. Number 136 is added to 5B7 and the sum obtained is 7A3, where A and B are integers. It is given that 7A3 is exactly divisible by 3. The only possible value of B is

(a) 2 (b) 5
(c) 7 (d) 8

Ans : (d)

$$\begin{array}{r} 1\ 3\ 6 \\ +\ 5\ B\ 7 \\ \hline 7\ 2\ 3 \end{array}$$

$\because$ Divisibility of 3 : The sum of digits must be divisible by 3
then, 7 A 3 $\Rightarrow$ 7 + A + 3
$\Rightarrow$ 10 + A
$\therefore$ $\boxed{A = 2}$
$\therefore$

$$\begin{array}{r} 1\ 3\ 6 \\ +\ 5\ B\ 7 \\ \hline 7\ 2\ 3 \end{array}$$

Hence, $\boxed{B = 8}$

77. A printer numbers the pages of a book starting with 1 used 3089 digits in all. How many pages does the book have ?

(a) 1040 (b) 1048
(c) 1049 (d) 1050

Ans : (c) Number of digits from 1 to 9 = 9 × 1 =9
Number of digits from 10 to 99 = 90 × 2 $\Rightarrow$ 180
Number of digits from 100 to 999 = 900 × 3 = 2700
Hence,
Total number of digits from 1 to 999 =9 +180 +2700 = 2889
For using 3089 digits,
remaining digits= 3089 – 2889 = 200
Now all the pages after 999 will use 4 digits
then number of pages with 4 digits = $\frac{200}{4}$ = 50 pages
Hence,
Total pages= 9 + 90 + 900 + 50 = 1049

78. How many triplets (x,y,z) satisfy the equation x+y+z=6, where x, y and z are natural numbers?

(a) 4 (b) 5
(c) 9 (d) 10

Ans : (d) From,
(permutation & combination)
$\Rightarrow {}^{(6-1)}C_{(6-4)}$
$\Rightarrow {}^{5}C_{2}$
$\Rightarrow \frac{\lfloor 5}{\lfloor 2 \lfloor 3}$ $\because {}^{n}C_{r} = \frac{\lfloor n}{\lfloor r \lfloor (n-r)}$
$= \frac{\lfloor 5}{\lfloor 2\times \lfloor 3} = \frac{5\times 4\times \lfloor 3}{2\times 1\times \lfloor 3} = 10$

79. Suppose you have sufficient amount of rupee currency in three denominations : ₹ 1, ₹ 10 and ₹ 50 . In how many different ways can you pay a bill of ₹ 107 ?

(a) 16 (b) 17
(c) 18 (d) 19

Ans : (c) Condition (I), when 50 rupee notes are not used :
(i) 0 ten rupee note + 107 ones = 107
(ii) ten rupee note + 97 one = 107
(iii) 2 ten rupee note + 87 ones = 107
: :
: :
(iv) 10 tens rupee note + 7 ones = 107
Possible ways =11
Condition : (ii)
When one fifty rupee note is used :
(i) o ten rupee note + 57 ones + 1 fifty = 107
(ii) 1 ten rupee note + 47 ones + 1 fifty = 107
(iii) 5 tens + 7 ones + 1 fifty = 107
Possible ways =6
Conditions (III)
When two fifty rupee notes are used :
$\Rightarrow$ (i) o ten + 7 ones + 2 fifty = 107
Possible ways =1
$\therefore$ Total possible ways = 11 + 6 + 1 = 18

80. Each face of a cube can be painted in black or white colours. In how many different ways can the cube be painted?

(a) 9 (b) 10
(c) 11 (d) 12

Ans : (b) White→ (W)
Black→(B)
Possible Ways-
(i) WW, WW, WW
(ii) WW, WW, WB
(iii)WW, WB, WB
(iv) WW, WW, BB
(v) WW, WB, BB
(vi) WB, WB, WB
(vii) WW, BB, BB
(viii) WB, WB, BB
(ix) WB, BB, BB
(x) BB, BB,BB
→ Possible ways = 10